Alibaba Cloud

Cross-border e-commerce for Southeast Asia with flash sales and an AI shopping assistant

Architecture proposal

Date	Revisions	Products	Catalog checked
1 November 2026	2	14	2026-10-31

Prepared with Architecture Builder

AI-generated. Review with a solutions architect before you build.

Summary

The design

I'll design a cross-border e-commerce platform for Southeast Asia with flash sale scaling and an AI shopping assistant. App servers auto-scale for flash sales, with Tair caching hot products and PolarDB as the main database. RocketMQ decouples order processing to absorb flash sale spikes without dropping requests. The AI shopping assistant uses Model Studio via AI Gateway to answer customer questions in local languages.

14 products, 11 connections, 2 revisions

Pinned requirements

No requirements were pinned.

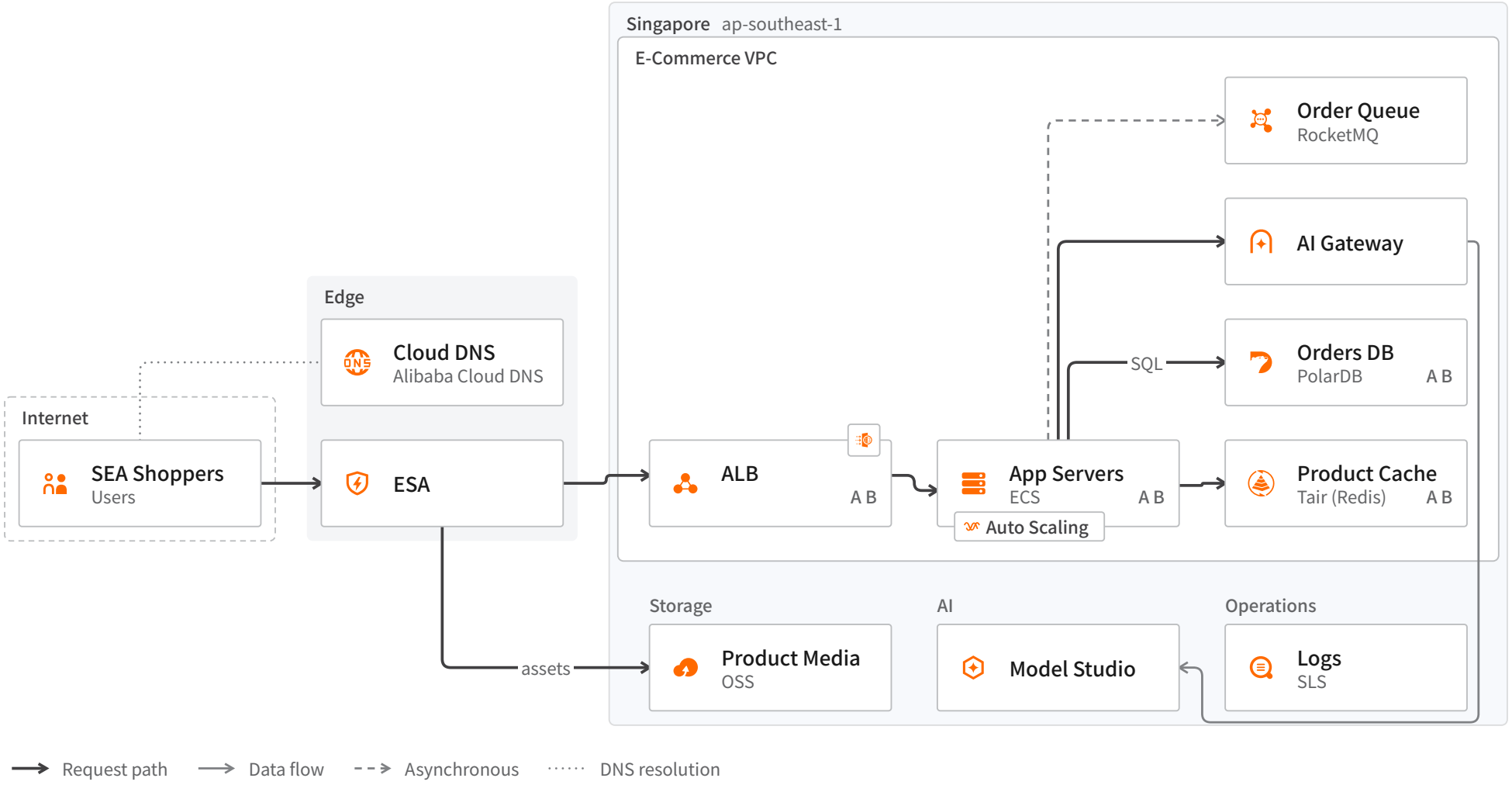
Constraints

No region or product constraints were set.

Assumptions

No assumptions were recorded.

Architecture diagram



Components

Every product in the diagram, why it is in this design, and where to read its official documentation.

Users

SEA Shoppers
Internet

Role in this design
Customers across Southeast Asia browsing and buying.

What it does
People reaching the system over the internet from browsers or apps.

Official documentation
Documentation link not yet available

Alibaba Cloud DNS

Edge · Domain Names and Websites

Role in this design
Resolves the shop domain to ESA edge nodes.

What it does
Authoritative public DNS for your domains; a resolve hop, not on the data path.

Official documentation
www.alibabacloud.com/product/dns

Edge Security Acceleration (ESA)

Edge · Networking & CDN

Role in this design
Accelerates dynamic content and caches assets across SEA.

What it does
Global edge: caching, dynamic acceleration, WAF, DDoS defense, bot management and edge functions.

Official documentation
www.alibabacloud.com/product/esa

Application Load Balancer (ALB)

Singapore (ap-southeast-1) · E-Commerce VPC · Zones A, B

Role in this design
Routes traffic to app servers across two zones.

What it does
Layer-7 HTTP/HTTPS/gRPC load balancer with content routing; also serves as ACK Ingress.

Official documentation
www.alibabacloud.com/product/server-load-balancer

Web Application Firewall (WAF)

Singapore (ap-southeast-1) · E-Commerce VPC · On ALB

Role in this design
Protects against bots and injection during flash sales.

What it does
WAF 3.0: blocks OWASP attacks, CC floods and bots via cloud-native or CNAME access.

Official documentation
www.alibabacloud.com/product/web-application-firewall

Elastic Compute Service (ECS)

App Servers
Singapore (ap-southeast-1) · E-Commerce VPC · Zones A, B

Role in this design
Runs the e-commerce application logic.

What it does
Virtual machines (x86, Arm, GPU) with cloud disks and security groups.

Official documentation
www.alibabacloud.com/product/ecs

Auto Scaling (ESS)

Singapore (ap-southeast-1) · E-Commerce VPC · On App Servers

Role in this design
Scales instances rapidly during flash sales.

What it does
Scales ECS or ECI fleets by rules, schedules or metrics and registers them to load balancers.

Official documentation
www.alibabacloud.com/product/auto-scaling

ApsaraMQ for RocketMQ

Order Queue
Singapore (ap-southeast-1) · E-Commerce VPC · Middleware

Role in this design
Buffers order events during flash sale traffic spikes.

What it does
Managed RocketMQ 5.x for transactional, scheduled and ordered messaging.

Official documentation
www.alibabacloud.com/product/mq

AI Gateway

Singapore (ap-southeast-1) · E-Commerce VPC · Middleware

Role in this design
Routes LLM calls with token limits and fallback.

What it does
LLM, MCP and agent gateway: multi-model routing, fallback, token limits, caching and guardrails.

Official documentation
www.alibabacloud.com/help/en/api-gateway/ai-gateway/product-overview/what-is-an-ai-gateway

PolarDB

Orders DB
Singapore (ap-southeast-1) · E-Commerce VPC · Zones A, B

Role in this design
Stores orders and inventory with instant read scale-out.

What it does
Cloud-native relational database (MySQL, PostgreSQL, Oracle-compatible) with compute-storage separation.

Official documentation
www.alibabacloud.com/product/polardb

Tair (Redis® OSS-Compatible)

Product Cache
Singapore (ap-southeast-1) · E-Commerce VPC · Zones A, B

Role in this design
Caches hot product data and sessions for flash sales.

What it does
Managed Redis-compatible in-memory database and cache, plus Tair enterprise features.

Official documentation
www.alibabacloud.com/product/tair

Object Storage Service (OSS)

Product Media
Singapore (ap-southeast-1) · regional services · Storage

Role in this design
Stores product images and videos served via ESA.

What it does
Durable object storage for files, media, backups and data lakes; S3-compatible.

Official documentation
www.alibabacloud.com/product/object-storage-service

Alibaba Cloud Model Studio

Singapore (ap-southeast-1) · regional services · AI & Machine Learning

Role in this design
Powers the AI shopping assistant with Qwen models.

What it does
Managed model APIs for Qwen (such as qwen3.8-max) and third-party models; OpenAI-compatible.

Price
Pay per token; qwen3.8-max $2 per 1M input and $6 per 1M output tokens (Singapore); 1M free tokens per model for 90 days, as of 2026-10-31

Official documentation
www.alibabacloud.com/product/modelstudio

Simple Log Service (SLS)

Logs
Singapore (ap-southeast-1) · regional services · Storage

Role in this design
Centralizes logs for monitoring flash sale performance.

What it does
Collects, searches, analyzes with SQL and alerts on logs, metrics and traces.

Official documentation
www.alibabacloud.com/product/log-service

Connections

How the products in the diagram talk to each other.

From	To	Carries	Type
SEA Shoppers	Cloud DNS		DNS resolution
SEA Shoppers	ESA	HTTPS	Data
ESA	ALB	origin	Data
ALB	App Servers		Data
App Servers	Product Cache		Data
App Servers	Orders DB	SQL	Data
App Servers	Order Queue		Asynchronous
ESA	Product Media	assets	Data
App Servers	AI Gateway		Data
AI Gateway	Model Studio		Data
App Servers	Logs		Control

Revision history

Every ask and what it changed in the design.

Revision 1

Drawn in 25.1 s

Ask

Cross-border e-commerce for Southeast Asia with flash sales and an AI shopping assistant

Changes

+ Users · SEA Shoppers · in Internet
+ Cloud DNS
+ ESA · in Edge
+ ALB · zones A, B
+ WAF · on ALB
+ ECS · App Servers · zones A, B
+ Auto Scaling · on ECS
+ Tair (Redis) · Product Cache · zones A, B
+ PolarDB · Orders DB · zones A, B
+ RocketMQ · Order Queue · in E-Commerce VPC
+ OSS · Product Media · in Singapore (ap-southeast-1)
+ Model Studio · in Singapore (ap-southeast-1)
+ AI Gateway · in E-Commerce VPC
+ SLS · Logs · in Singapore (ap-southeast-1)
+ 11 connections

Revision 2

Drawn in 3.6 s

Ask

oh i also want waf

Changes

No change to the diagram.